FODS ASSIGNMENT 4

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Batch – 2	
DATE – 04-08-2026	TIME – 8:00 TO 10:00

QUESTION) Practical 4: Polynomial Representation Using Arrays

Represent single-variable polynomials using arrays or structures. Implement polynomial addition and multiplication using array representation.

Answer:

ALGORITHM:

ALGORITHM FOR POLYNOMIAL ADDITION:

Step1] Start.
Step2] Read the highest degree of the first polynomial.
Step3] Read the coefficients of the first polynomial from highest degree to constant term.
Step3] Read the highest degree of the second polynomial.
Step4] Read the coefficients of the second polynomial from highest degree to constant term.
Step5] Find the maximum degree of both polynomials.
Step6] For each exponent from **0 to maximum degree**:

Step7] Add the corresponding coefficients of both polynomials.

Step8] Store the result in the sum array.

Step9] Display the resultant polynomial.
Step10] Stop.

ALGORITHM FOR POLYNOMIAL MULTIPLICATION:

Step1] Start.

Step2] Read the highest degree and coefficients of the first polynomial.

Step3] Read the highest degree and coefficients of the second polynomial.

Step4] Initialize all elements of the product array to 0.

Step5] For each coefficient of the first polynomial:

Step6] For each coefficient of the second polynomial:

- Multiply the two coefficients.
- Store the result in the position corresponding to the sum of their exponents.

Step7] Display the resultant polynomial.

Step8] Stop.

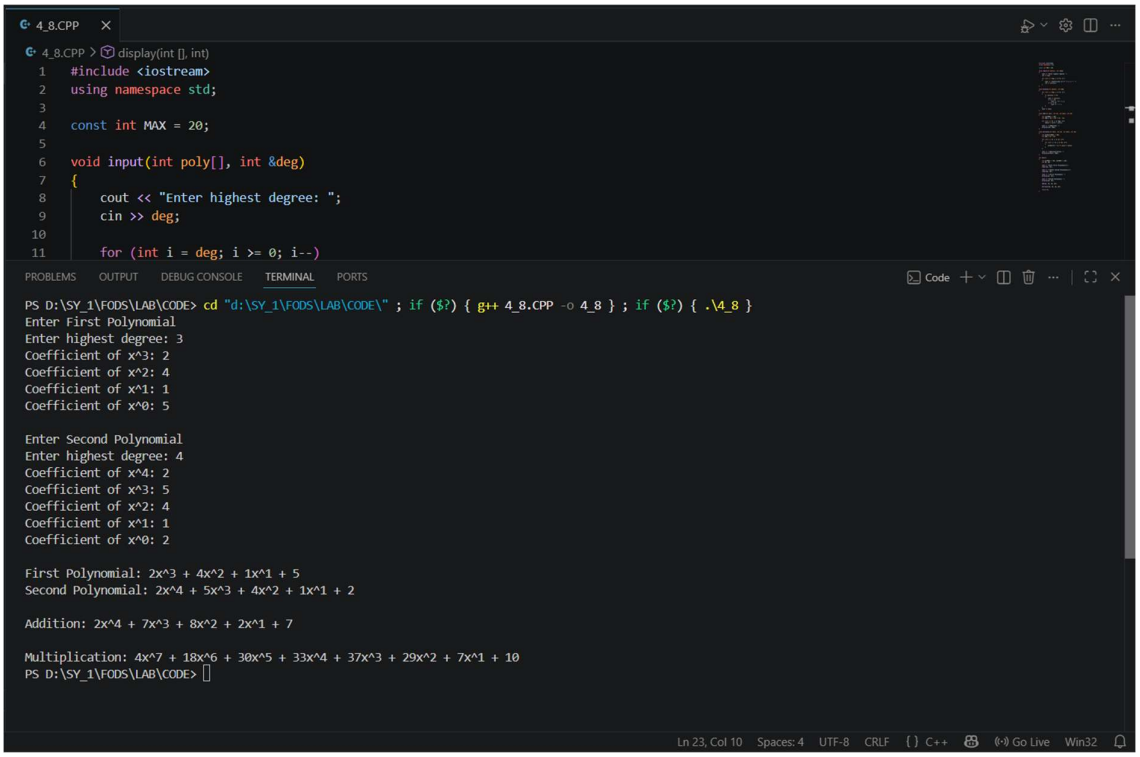
CODE:

```
#include <iostream>
using namespace std;
const int MAX = 20;
void input(int poly[], int &deg)
{
   cout << "Enter highest degree: ";
   cin >> deg;
 for (int i = deg; i >= 0; i--)
   {
      cout << "Coefficient of x^" << i << ": ";
      cin >> poly[i];
   }
}
void display(int poly[], int deg)
{
   for (int i = deg; i >= 0; i--)
   {
      if (poly[i] != 0)
```

```
      {
         cout << poly[i];
         if (i > 0)
            cout << "x^" << i;
         if (i != 0)
            cout << " + ";
      }
   }
   cout << endl;
}
void add(int p1[], int d1, int p2[], int d2)
{
   int sum[MAX] = {0};
   int deg = (d1 > d2) ? d1 : d2;
for (int i = 0; i <= deg; i++)
      sum[i] = p1[i] + p2[i];
  cout << "\nAddition: ";
   display(sum, deg);
}
void multiply(int p1[], int d1, int p2[], int d2)
{
   int product[MAX] = {0};
 int deg = d1 + d2;
for (int i = 0; i <= d1; i++)
   {
      for (int j = 0; j <= d2; j++)
      {
         product[i + j] += p1[i] * p2[j];
      }
   }
```

```
 cout << "\nMultiplication: ";
    display(product, deg);
}
int main()
{
    int p1[MAX] = {0}, p2[MAX] = {0};
    int d1, d2;
 cout << "Enter First Polynomial\n";
    input(p1, d1);
cout << "\nEnter Second Polynomial\n";
    input(p2, d2);
 cout << "\nFirst Polynomial: ";
    display(p1, d1);
 cout << "Second Polynomial: ";
    display(p2, d2);
 add(p1, d1, p2, d2);
 multiply(p1, d1, p2, d2);
  return 0;
}
```

OUTPUT:


4_8.CPP
4_8.CPP > display(int [], int)
#include <iostream>
using namespace std;
const int MAX = 20;
void input(int poly[], int &deg)
{
cout << "Enter highest degree: ";
cin >> deg;
for (int i = deg; i >= 0; i--)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
PS D:\SY_1\FODS\LAB\CODE> cd "d:\SY_1\FODS\LAB\CODE\" ; if ($?) { g++ 4_8.CPP -o 4_8 } ; if ($?) { .\4_8 }
Enter First Polynomial
Enter highest degree: 3
Coefficient of x^3: 2
Coefficient of x^2: 4
Coefficient of x^1: 1
Coefficient of x^0: 5
Enter Second Polynomial
Enter highest degree: 4
Coefficient of x^4: 2
Coefficient of x^3: 5
Coefficient of x^2: 4
Coefficient of x^1: 1
Coefficient of x^0: 2
First Polynomial: 2x^3 + 4x^2 + 1x^1 + 5
Second Polynomial: 2x^4 + 5x^3 + 4x^2 + 1x^1 + 2
Addition: 2x^4 + 7x^3 + 8x^2 + 2x^1 + 7
Multiplication: 4x^7 + 18x^6 + 30x^5 + 33x^4 + 37x^3 + 29x^2 + 7x^1 + 10
PS D:\SY_1\FODS\LAB\CODE>
Ln 23, Col 10 Spaces: 4 UTF-8 CRLF { } C++ Go Live Win32